

Hal Hoffmeyer

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EDUCATION

University of Michigan | B.S.E. Computer Science | GPA: 3.97 May 2027

Coursework: Distributed Systems, Machine Learning, Web Systems, Computer Organization, Data Structures and Algorithms, Discrete Math, Introduction to Probability, Applied Linear Algebra, Statistics and Data Analysis

Awards: James B. Angell Scholar, William J. Branstrom Freshman Prize, University Honors

EXPERIENCE

Zeta Pi Tech Fraternity | Full-Stack Web Developer June 2025 — Present

- Created ZProfile, a full-stack Next.js web app for 100+ users designed to streamline management and logistics.
- Implemented an attendance tracker and event creator integrated with Supabase using a custom dual list box component and React Hook Form.
- Built profile page allowing students to log course history by pulling course information from U-M API and writing custom multiselect component to store student course data into Supabase.
- Developed an interactive course directory by querying previously stored Supabase data and built logic for search, debounce, and filtering features using React hooks and PostgreSQL.
- Implemented a bug reporting system leveraging the GitHub REST API, automatically creating GitHub issues from user submissions to streamline bug tracking and improve response time.

Detroit Techno Film Web Platform | Web Developer September 2024 — October 2025

- Developed a companion web app for a documentary, using Next.js to showcase dynamic music and video dashboards, NextAuth.js for secure user authentication, AWS S3 for video storage, and Vercel for database.
- Led developer meetings to establish a unified Git workflow amongst team members and merge back-end with front-end components.

Kumon | Tutor June 2023 — August 2024

- Provided online and in-person tutoring in math and English for K-12 students and graded homework.

PROJECTS

[MNIST Classifier](#) | C++ March 2026

- Developed a neural network from scratch for MNIST digit classification, achieving 98% accuracy using ReLU and Softmax activation functions, optimized forward and back propagation, and hyperparameter sweeping.

ICU Patient Mortality Prediction | Python, NumPy, Scikit-learn, Pandas February 2026

- Developed a Logistic Regression classifier to predict ICU patient mortality, implementing L1/L2 regularization and extensive hyperparameter sweeps to optimize model selection.
- Engineered a high-dimensional feature pipeline from irregular ICU time-series data, utilizing 5-fold cross-validation and bootstrap sampling to achieve an AUROC of 0.85 on unseen clinical test data.

Sharded Key/Value Store with Paxos Groups | Go April 2026

- Built a fault-tolerant, sharded key/value store in Go with shard migration across Paxos replicated groups.
- Implemented a shard master for dynamic shard rebalancing, atomic state handoffs, and duplicate request detection. Ensured linearizability by serializing all operations in the Paxos consensus log.

[Minesweeper](#) | React, TypeScript, Tailwind December 2024

- Developed a Minesweeper game, integrating game logic while handling state management and dynamic grid rendering with React components and event listeners in TypeScript for a seamless user experience.

Assembly Simulator | Assembly, C September 2025

- Engineered a program in assembly that executes 16-bit integer multiplication using a custom algorithm, without the use of bit-shift and multiply instructions. Compiled and simulated with a separate C program.

Languages: C/C++, JavaScript/TypeScript, HTML/CSS, SQL, C#, Go, Python, R, Java

Tools: React, jQuery, .NET, PostgreSQL, NumPy, Git, SVN, Next.js, Jira, Makefile, Vim, VS Code, Visual Studio, IntelliJ

Certifications: N3 Japanese Language Proficiency Test